Smartphone-based anaemia detection using eye imaging

"in the smart phone comaptible machine learing technology where the image of the eye is processedd and the result is given as anaemic or not" *Generated by LeapSpace*

Yes — that is a real smartphone-based eye-image anaemia screening approach.

Several studies describe **smartphone-captured eye or conjunctival images** being processed by machine learning to classify a person as **anaemic or non-anaemic**. The closest match to your description is **conjunctiva/palpebral conjunctiva imaging**, where the phone camera captures the eye region and the model outputs an anaemia result or haemoglobin estimate [57] [58] [59] [60].

What this technology does

- **Input:** a smartphone photo or video of the **eye/conjunctiva** [57] [60] [61].
- **Processing:** image segmentation or feature extraction from the conjunctival area, then classification with ML/DL models such as CNNs, U-Net + CNN, Random Forest, SVM, or hybrid pipelines [57] [59] [62] [63].
- **Output:** either
  - a **binary result**: anaemic vs non-anaemic [62] [63], or
  - a **haemoglobin estimate** that is then used to decide anaemia status [57] [60] [61].

How well it works

- A retinal fundus study showed that anaemia can be detected from **eye images plus metadata**, with the combined model performing better than image-only or metadata-only models [9].
- A conjunctival-image study using smartphone photos reported **high specificity but low sensitivity** in one dataset, showing that performance can vary by population and image quality [57].
- Another smartphone-based conjunctiva system reported **strong predictive accuracy** and was designed for real-time screening [60].
- Broader reviews and clinical studies note that **smartphone ocular imaging is promising**, but real-world performance depends on lighting, pigmentation, device differences, and external validation [9] [64] [65] [66].

Bottom line

If you mean **"a phone takes an eye photo, AI analyses it, and tells whether the person is anaemic"**, then the term you're looking for is **smartphone-based conjunctival/retinal anaemia screening using machine learning** [9] [60] [61]. It exists and is promising, but the evidence shows that **conjunctiva-based screening is more variable than blood-test-based ML and still needs stronger real-world validation** [9] [57] [65] [66].

Confidence level: **High** — multiple independent studies directly describe smartphone eye-image anaemia detection, including conjunctiva-based binary classification and haemoglobin prediction [57] [60] [61] [66].

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